Project Report

Medical Recommendation System with Machine Learning

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1. Introduction

1.1 Background

Healthcare is one of the fastest-growing and most critical domains globally. Despite advancements, millions of people continue to face challenges such as limited access to doctors, misdiagnosis, lack of health awareness, and overcrowded hospitals. With the rapid growth of technology, **Artificial Intelligence (AI) and Machine Learning (ML)** have emerged as powerful tools to address healthcare challenges.

An intelligent **Medical Recommendation System** can act as a **virtual health assistant**, predicting diseases from patient symptoms and recommending medications, diets, precautions, and workouts. This provides patients with **early awareness**, helps them make informed decisions, and reduces dependency on unreliable internet sources.

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2. Problem Statement

2.1 Background of the Problem

- Patients often search online for their symptoms and end up with misinformation.
- Rural areas suffer from a lack of medical professionals, forcing people to travel long distances for basic consultations.
- Hospitals and clinics are overcrowded, leading to delayed diagnosis and treatment.
- Preventive healthcare is often ignored due to lack of awareness.

2.2 Core Problem

There is no easily accessible, intelligent, and reliable system for **predicting diseases based on symptoms** and **providing safe medical recommendations**.

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3. Proposed Solution

The proposed solution is an **AI-powered Medical Recommendation System** that:

1. Takes patient **symptoms** as input.

2. Uses **Machine Learning models** trained on medical datasets to predict possible diseases.
3. Recommends:
    - **Medications** (general safe medicines).
    - **Diets** (nutritional guidance).
    - **Precautions** (preventive steps).
    - **Workouts** (fitness routines for health improvement).
4. Provides a **user-friendly web application** interface using Streamlit for real-time interaction.

This ensures patients receive **early health guidance**, while still encouraging them to consult certified doctors for treatment.

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4. Project Objectives

- Develop a reliable ML model for **disease prediction**.
- Create a **recommendation system** for medications, diets, precautions, and workouts.
- Design a **web-based application** for easy use by patients.
- Promote **preventive healthcare** through personalized advice.
- Bridge the healthcare gap in **rural and underserved communities**.

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5. Literature Review

Several researchers have attempted AI-based healthcare systems:

- Disease prediction using **Decision Tree** and **Random Forest** models has shown high accuracy in structured datasets.
- ML-based health chatbots provide symptom-based predictions but lack **lifestyle recommendations** (diet/workout).
- Our project improves existing models by integrating **disease prediction + complete medical guidance** in a **single platform**.

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6. Methodology

6.1 Dataset Used

- **Training.csv** → Symptoms and diseases.
- **Symptom-severity.csv** → Severity levels of symptoms.
- **medications.csv** → General medicines for diseases.
- **diets.csv** → Dietary recommendations.
- **precautions_df.csv** → Precautions to take.
- **workout_df.csv** → Workouts for healthy recovery.

6.2 Preprocessing Steps

- Data cleaning (removing missing/duplicate values).
- Converting categorical symptoms into numerical form.
- Assigning severity scores for accurate prediction.
- Splitting data into training and testing sets.

6.3 Machine Learning Models Used

- **Decision Tree Classifier**
- **Random Forest Classifier**
- **Support Vector Machine (SVM)**

👉 Out of these, the **best-performing model** was selected based on accuracy.

6.4 System Architecture

User → Input Symptoms → ML Model → Disease Prediction → Recommendation System → Output (Medications, Diet, Precautions, Workouts)

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7. Tools and Technologies Used

- **Programming Language**: Python
- **Libraries**: Pandas, NumPy, Scikit-learn, Streamlit, Pickle
- **IDE**: Jupyter Notebook, VS Code
- **Framework**: Streamlit (for Web App)
- **Version Control**: Git/GitHub (optional)

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8. Implementation

8.1 Model Training

- ML model trained using **Training.csv**.
- Used **train-test split** for validation.
- Saved best model as svc.pkl using Pickle.

8.2 Web Application (Streamlit)

- Sidebar for navigation (Landing Page, About, Prediction).
- **Landing Page** → Project title, quote, and image.
- **About Section** → Features, project details, benefits.
- **Prediction Section** → Users enter symptoms → Predicted disease + Recommendations displayed.

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9. Results

- Achieved **high accuracy** in disease prediction (85–95% depending on model).
- Successfully integrated recommendations (medications, diets, workouts, precautions).
- Developed a clean and interactive Streamlit web app.

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10. Expected Outcome

- A functional **AI-powered medical recommendation system**.
- Accessible healthcare assistance for people in **rural and semi-urban areas**.
- Increased health awareness through **personalized recommendations**.
- Encouragement of **preventive healthcare**.

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11. Impact

- **Patients**: Early awareness and guidance.
- **Doctors**: Can be used as a **support system** for diagnosis.
- **Society**: Reduced dependency on unreliable online sources.
- **Healthcare System**: Helps reduce burden on hospitals for minor health issues.

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12. Limitations

- Recommendations are **general** and not a substitute for doctor consultation.
- Limited dataset size; adding more medical data would improve accuracy.
- Cannot handle **emergency medical situations**.

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13. Future Enhancements

- Integration with **Electronic Health Records (EHRs)**.
- Adding **real-time chatbots** for interaction.
- Multi-language support for rural users.
- Mobile app version for wider accessibility.
- Integration with wearable devices (Fitbit, Smartwatches) for real-time health monitoring.

14. Conclusion

The **Medical Recommendation System** successfully predicts diseases from symptoms and provides safe, general recommendations on medications, diet, workouts, and precautions.
It acts as a **virtual health assistant**, supporting patients in making informed health decisions and bridging the gap in healthcare accessibility.
While it cannot replace certified doctors, it ensures **awareness, preventive care, and early intervention**, which can significantly improve healthcare outcomes.